

Article

Preference-Aware Multimodal Journey Planner: An Optimization Approach for Smart Mobility

Bia Mandžuka ^{1,*}, Krešimir Vidović ¹ , Marko Ševrović ¹  and Jasmin Čelić ² 

¹ Faculty of Transport and Traffic Sciences, University of Zagreb, 10000 Zagreb, Croatia; kvidovic@fpz.unizg.hr (K.V.); msevrovic@fpz.unizg.hr (M.Š.)

² Faculty of Maritime Studies, University of Rijeka, 51000 Rijeka, Croatia; jasmin.celic@pfri.uniri.hr

* Correspondence: bmandzuka@fpz.unizg.hr

Highlights

This paper presents a methodological proof-of-concept based on an inverse-optimization approach for discovering individual traveller preferences and applies it to multimodal journey planning in the context of smart mobility. Within the available methodological testbed, the preference-discovery-based model achieved closer in-sample agreement with user-provided route evaluations than the model based on explicitly rated criteria. This finding highlights the limitations of explicit user-defined criterion weighting, which is still commonly used in current journey planners and supports preference discovery as a more appropriate basis for the initial setup of user-specific route-choice preferences. Beyond personalization, the model opens a pathway toward incorporating broader goals of sustainable urban mobility, including system-aware re-ranking of route alternatives.

What are the main findings?

- The proposed preference-discovery-based model achieved closer in-sample agreement with user-provided route evaluations than the model based on explicitly rated criteria, highlighting the limitations of explicit user-defined criterion weighting in current journey planners and supporting preference discovery as a more suitable basis for initial user-preference setup.
- The proposed preference-discovery-based model, built on an inverse decision-making model, provides a transparent and interpretable framework for the initial setup of user-specific preferences in multimodal route ranking.

What are the implications of the main findings?

- Personalization in Multimodal Journey Planners (MJPs) should not be based on generic or explicitly specified criterion weights, but on numerically estimated individual user preferences.
- MJPs can be developed not only as tools for supporting individual users, but also as transparent mechanisms for aligning individual choices with broader goals of sustainable urban mobility.



Academic Editors: Pierluigi Siano and Frank Witlox

Received: 29 April 2026

Revised: 13 June 2026

Accepted: 16 June 2026

Published: 19 June 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

Abstract

This paper examines the role of Multimodal Journey Planners (MJPs) as a link between user-oriented personalization and the broader societal goals of sustainable urban mobility. In smart cities, MJPs may serve as digital decision-support tools that connect individual mobility choices with broader sustainability objectives. Although contemporary journey planners increasingly display multiple criteria, such as travel time, cost, CO₂ emissions, and number of transfers, they still generally rely on predefined and non-personalized criterion weights and rarely infer travellers' actual preferences from observed choices. The

paper therefore proposes a transparent methodological proof-of-concept that combines multicriteria decision-making and inverse optimization to discover individual preference weights and enable personalized, preference-aware planning of multimodal routes. The Weighted Sum Method (WSM) is adopted as the basic ranking framework, and the proposed approach is evaluated within a controlled methodological testbed based on multimodal journey scenarios in Vienna. The results indicate that, within the available methodological testbed, the preference-discovery-based model achieved closer in-sample agreement with user-provided route evaluations than the model based on explicitly rated criteria. This was observed in the ranking-agreement analysis, where a more favourable penalty-point ratio was obtained in 19/21 cases (90.5%) and in the numerical error comparison, where lower in-sample reconstruction errors were obtained for 18/21 users (85.71%) across all scenarios. The paper further considers the tension between individual and system-level goals, as well as a conceptual extension toward system-aware re-ranking of alternatives. Within the broader framework of smart mobility, the importance of interoperability and open data is also recognized, with National Access Points (NAPs) for multimodal travel information potentially representing an important precondition for the development of advanced and transparent MJP solutions.

Keywords: multimodal travel; multimodal journey planners; preference discovery; multicriteria decision-making; inverse optimization; personalization; smart mobility; sustainable urban mobility

1. Introduction

Multimodal travel involves a combination of several transport modes within a single trip and requires users to understand and coordinate multiple interdependent journey segments [1]. For that reason, the choice of a multimodal route is not merely a technical problem of navigation, but also a complex decision-making process in which the user simultaneously evaluates multiple travel attributes, such as time, number of transfers, cost, effort, uncertainty, and perceived safety. Compared with unimodal travel, and especially with the use of a private car, multimodal travel is less intuitive and cognitively more demanding for the user [2–5].

Intelligent mobility is a key component of the smart city concept, as it enables more efficient, safer, and more sustainable movement of people and goods. The application of advanced technologies, such as information and communication technologies, artificial intelligence, and real-time traffic management systems, can reduce traffic congestion, emissions, and, ultimately, travel times. In addition, the integration of different modes of transport, that is, multimodality, increases accessibility and quality of life for citizens. Intelligent mobility also contributes to better infrastructure planning and data-driven decision-making, which is essential for the long-term sustainable development of urban areas.

Under such circumstances, information support plays a crucial role by linking the complex transport system and the end user [6]. This is precisely why Multimodal Journey Planners (MJPs) are increasingly viewed as an important component of smart mobility and of the broader smart city framework [7,8]. More broadly, smart urban mobility research increasingly emphasizes the role of information technologies, data-driven approaches, and Intelligent Transport Systems (ITSs) in addressing urban mobility challenges [9,10]. Their role is not merely to provide information on possible routes, but also to transform the complexity of a multimodal transport system into an operationally usable and user-

understandable decision. In this way, MJPs become not only navigation tools, but also digital decision-support mechanisms within urban mobility systems.

In this context, MaaS-oriented services further emphasize the practical relevance of multimodal journey planning. MaaS aims to integrate multiple heterogeneous transport modes into a single user-oriented service, enabling travellers to plan itineraries, access different mobility services and manage booking or payment through a common platform. Therefore, MJPs represent an important operational component for supporting seamless and user-oriented multimodal travel within smart mobility systems [11].

However, most existing journey planners are still predominantly based on the logic of the shortest route or the shortest travel time, with an implicit assumption of universal optimality. Although contemporary systems increasingly display multiple criteria, their logic of ranking alternatives still generally relies on predefined, generic, and non-personalized criterion weights [12,13]. In the context of multimodal travel, such an approach is insufficient because users do not value all travel attributes equally; rather, their decisions depend on individual preferences, experience, transfer tolerance, perceived effort, and other contextual factors [14,15].

This is where the main research gap can be identified. Despite growing interest in multimodal journey planning, personalized passenger information systems, and preference-aware recommendation, there is still a lack of an approach that would simultaneously account for the multicriteria nature of multimodal route choice, the heterogeneity of user preferences, and the limitations of explicit preference elicitation among non-expert users [6,16–18]. Existing approaches that attempt to incorporate user preferences often rely on the direct specification of criterion importance, which may be cognitively demanding and may lead to inconsistent results [12].

Building on the above, the primary aim of this paper is to develop a multimodal travel choice model based on travellers' preferences, as discovered through the analysis of their previous choices, while also incorporating broader optimization goals at the level of the urban mobility system. In this sense, the paper is based on the assumption that conclusions about individual user preferences can be drawn from the evaluation of users' choices among alternative specific journeys, and that such implicitly discovered preferences can be used as the basis for the personalized ranking of a proposed set of multimodal routes.

The main contributions of the paper can be summarized as follows. First, it conceptually links multimodal journey planning, Recommender Systems (RSs), and multicriteria decision-making within a preference-aware framework relevant to smart mobility. Second, it demonstrates the limitations of approaches based on the explicit specification of criterion importance by non-expert users. Third, it proposes a methodology for discovering user preferences based on an inverse decision-making model, through which individual criterion weights are estimated implicitly from user evaluations of the offered alternatives. Fourth, it develops a multimodal route choice model based on such discovered preferences, with the Weighted Sum Method (WSM) selected as the basic ranking framework because of its transparency, interpretability, and suitability for operational implementation. Finally, the paper also considers a conceptual extension of the model toward system-aware re-ranking of alternatives, thereby opening up the possibility of incorporating broader transport-system goals into the final recommendation step. Within the broader framework of the digital and data infrastructure of smart mobility, the paper additionally recognizes the importance of interoperability and open data, with National Access Points (NAPs) for multimodal travel information representing a potentially important institutional and technical precondition for the development of advanced and personalized MJP solutions [19,20].

The remainder of the paper is structured as follows. Section 2 discusses the conceptual background and the research gap, with an emphasis on the complexity of multimodal

travel, the role of MJPs, and the need for preference-aware decision support in the context of smart mobility. Section 3 presents the methodological framework of the paper, including the choice of the multicriteria decision-making method, the preference-discovery model, the normalization procedure, and the conceptual extension toward system-aware re-ranking. Section 4 describes the case study and the evaluation design, while Section 5 presents the results of two complementary in-sample evaluation procedures: a ranking-agreement analysis and a numerical error comparison.

2. Conceptual Background

2.1. Multimodal Travel Complexity and the Need for Information Support

According to Van Nes and Bovy [1], multimodal travel is defined as a combination of several travel modes between an origin and a destination that may potentially be beneficial both for the traveller and for society. In recent EU transport policy, door-to-door multimodal travel has been increasingly emphasized as an important direction for improving passenger mobility through more integrated, seamless, and user-oriented transport services [21,22].

Multimodal travel involves the combination of several transport modes, for example, tram + bus + e-scooter, within a single OD relation. In this sense, the user (traveller) must understand and follow a route composed of multiple mutually dependent segments. In addition, today's multimodal systems are not fully harmonized and integrated, since they have developed gradually, over different periods and under the influence of different stakeholders, such as cities, operators, regulators, and private service providers [23,24]. This complexity is directly reflected in travellers' multimodal behaviour, that is, in the way they perceive and choose different combinations of transport.

From the traveller's perspective, the complexity of multimodal travel can be conceptualized through three interrelated dimensions: (1) cognitive load, (2) risk and uncertainty, including perceived safety during transfers, and (3) perceived transfer effort. Multimodal travel systems require the traveller to understand the segments of a multimodal route, monitor arrival and departure times, plan transfers, and coordinate multiple journey elements. In this sense, the more modes and transfers a journey involves, the greater the mental effort required to manage it [2,3]. Complexity is further increased by the perceived level of uncertainty, especially in relation to the risk of missing a connection, delay-related uncertainty, and similar factors. Multimodal journeys are therefore not complex solely because of the number of modes involved, but because of the combination of cognitive load, the sequential interdependence of route segments, and the perceived effort associated with transfers. Such complexity is not merely a practical problem, but an inherent characteristic of multimodal systems. Under such conditions, users still rely heavily on private cars, which are often perceived as a simpler, more comfortable, and more convenient form of mobility, that is, as an option that offers a less fragmented and more seamless travel experience [4,5]. Compared with a private car, multimodal travel is less intuitive and cognitively more demanding because it involves multiple travel *criteria*, the need to trade them off against one another, and the management of uncertainty.

Accordingly, information support plays a key role in linking the multimodal transport system and the user [3]. It enables multimodal travel, despite its inherent complexity, to become a coherent whole that is easier for the user to understand and use. Without adequate information support, the potential of multimodal systems may remain underutilized, even when infrastructure and the diversity of available modes are relatively well developed. This confirms that the operational usability of such systems depends not only on the quality of infrastructure and the organization of public transport, but also on the quality and relevance of information support [6]. In this sense, multimodal journey planning can also be viewed within the broader framework of smart cities, in which digital and information solutions

play an important role in increasing the functionality and acceptability of sustainable urban mobility. In this context, information support is provided through MJPs, which help users understand, compare, and choose appropriate multimodal travel options. Moreover, the function of MJPs is not limited to informing users but also includes guiding their choices towards more sustainable mobility patterns.

2.2. Multimodal Journey Planners as Interfaces Between Users and Transport Systems in Smart Mobility

In today's mobility, journey planning has evolved from simple route selection into a multicriteria process that encompasses a wide range of transport modes and information sources [25]. In such a context, MJPs play a key role as an intermediary component between users and complex transport systems. As part of Intelligent Transport Systems (ITSs), they provide information support to travellers in planning journeys from point A to point B by combining multiple transport modes. MJPs should therefore be understood not merely as digital navigation tools, but also as an important component of smart mobility, as they enable users to make operational use of a complex multimodal system [7].

As noted earlier, multimodal transport systems are inherently complex because they involve multiple travel modes, operators, fare systems, schedules, and infrastructural constraints. At the user level, this gives rise to a key problem, since journey choice requires the evaluation of many alternatives while simultaneously considering multiple criteria, such as travel time, number of transfers, and travel cost. Such cognitive overload, in which travellers must process large amounts of data and integrate multiple sources for increasingly complex planning tasks, represents a significant challenge in modern transport systems [6]. The role of MJPs is therefore to reduce that complexity to a level that users can cognitively process, while at the same time ensuring the functionality and acceptability of the proposed routes.

For such an information bridge to be effective, it is not enough merely to provide large volumes of data. Information support must be functional for the end user and must minimize the effort involved in route choice. A system that offers many routes but does not help the user identify which option is the most appropriate for them does not achieve its full value as a decision-support tool.

This is precisely why personalization becomes a key characteristic of multimodal journey planning tools [6]. It is not merely an additional functionality, but a necessary precondition for effective decision support. Users differ in their preferences and priorities, which means that an optimal route is not optimal for every user.

Despite technological advances and the availability of data from different sources, such as IoT sensor data, public transport data, and user GPS traces, many existing MJP implementations still rely on simplified assumptions about users. These tools often offer multiple-choice criteria, but at the same time assume fixed relationships among them or use predefined criterion weights that are not adapted to users at the individual level. Such an approach implicitly rests on the idea of *the average user*. This results in generic solutions that often fail to meet the specific needs and preferences of travellers, which may lead to frustration and reduced system use [14]. Empirical studies confirm a pronounced heterogeneity of traveller preferences. Research [15] indicates that transport mode choice is a multidimensional and context-dependent process that depends on a combination of users' socio-demographic characteristics, journey characteristics, transport-system performance, and region-specific circumstances.

2.3. Preference-Aware Travel Decision Support in Smart Mobility

Smart cities represent a transformative paradigm and an integrated framework for managing urban environments, within which smart mobility constitutes one of the central

components [26]. In this sense, the development of smart mobility implies the establishment of information-support systems capable of adapting information and recommendations to the individual needs of today's users. In the context of smart cities [8,27], Recommender Systems (RSs) represent an important digital component. Such systems play an important role in filtering information, supporting decision-making, and providing personalized services. Applications in mobility are particularly important because of their potential to suggest alternative routes, encourage the use of public transport, and support shared-mobility services.

Personalization in the context of smart mobility is not merely a matter of user interface design or aesthetic customization, but a fundamental conceptual assumption of systems adapted to user needs and preferences. According to Liu et al. [28], personalized travel recommendations affect user behaviour and also show potential for shaping sustainable traveller behaviour through so-called digital nudging [29]. Recent research in smart mobility points to a systematic shift and growing interest in the development of personalized Passenger Information Systems (PIs) [16–18]. Indeed, Van Ardenne et al. [6] propose a taxonomy comprising five levels of personalization, ranging from level 0 (no personalization) to level 4 (full personalization). According to this taxonomy, systems differ in the degree of automation and in the level of user interaction with the journey-planning system. Moreover, this need becomes particularly important in the context of Mobility-as-a-Service (MaaS), where personalization is recognized as a key precondition for encouraging changes in travel behaviour and increasing the acceptance of multimodal mobility [17]. Personalization therefore emerges as a necessary condition of smart mobility. This is in line with recent work emphasizing that, within the smart city paradigm, route-planning services should increasingly reflect the actual needs and preferences of end users [30].

This research stems from the fact that user mobility is not homogeneous but marked by significant variability in needs and preferences. Studies [13,15,31,32] show that there is no single average user, but rather several clearly differentiated groups with different mode-use patterns. It is precisely for this reason that approaches that account for user preferences in the choice of alternatives, that is, preference-aware approaches, become necessary. Such heterogeneity implies that the system must evaluate alternative routes according to individually relevant criteria.

Journey planning in contemporary transport systems cannot be viewed solely as a technical navigation problem, but as a decision-making process involving the evaluation of alternatives and trade-offs between often conflicting criteria, such as travel time and number of transfers [12]. From this perspective, multimodal route choice can be understood within the framework of multicriteria decision-making. Since multimodal journey planners generate and rank alternative travel options according to certain criteria, they may also be viewed as a specialized class of recommender systems (RSs), whose basic function is to facilitate decision-making. In this sense, the role of such systems is not merely to inform users, but to provide structured decision support in route choice among multiple possible alternatives.

The literature [16,18,33,34] indicates that MJPs are increasingly conceptualized as RSs, especially in the context of personalized ranking of alternative travel options, where alternatives are generated and ranked according to user preferences such as cost, travel time, and selected transport mode.

In this context, a so-called preference-aware approach is emerging, according to which the recommendation system does not assume a single optimal solution for all users, but instead ranks alternatives on the basis of user-specific preferences, while simultaneously accounting for the current state of the urban system and constraints defined by city authorities [17,35]. This is especially important in the context of MaaS systems, where different

transport options are integrated within a single service, making the need for preference-aware decision support even more pronounced. Preference-aware decision support thus becomes an important conceptual foundation of smart mobility systems that seek to reconcile the diversity of user preferences with the complexity of the multimodal transport environment. Such a preference-aware approach also opens up the broader question of the relationship between individual user preferences and the goals of the transport system, since a route that is individually optimal does not necessarily have to be optimal from the system perspective. In the remainder of the paper, this issue serves as a starting point for considering the possibility of aligning the user and system perspectives in the ranking of multimodal routes.

3. Methodology

3.1. Problem Definition

The choice of a multimodal route represents a complex multicriteria decision-making problem in which the user must simultaneously consider several travel attributes (criteria) [12]. Unlike unimodal travel, where the decision is often reduced to the optimization of one dominant criterion, such as travel time, multimodal travel involves a combination of several transport modes as well as additional elements such as the number of transfers, walking time, waiting time, perceived safety, and other route characteristics. For that reason, users implicitly evaluate trade-offs among several travel criteria when selecting a route.

Under such conditions, each alternative represents a possible multimodal route, while the criteria represent measurable travel attributes describing the quality of a particular route. Furthermore, considering the traveller's perception and behavioural characteristics, it is evident that the same multimodal route will not have the same value for all users. Since the perceived importance of individual travel criteria differs among travellers, the optimal route is not universal but depends on the traveller's individual preferences and travel "genome" [36].

Because the criteria are often mutually conflicting, users must implicitly establish a balance among different travel aspects when deciding. For example, a shorter overall travel time may be associated with a larger number of transfers or longer walking time. It is precisely for this reason that multimodal route choice cannot be reduced to the optimization of a single criterion only.

Although commercial journey planners increasingly include additional filters and contextual data, their basic recommendation logic is still typically based on travel-time efficiency and related operational criteria rather than on identified individual user preferences. In the context of multimodal travel, such an approach is insufficient because users simultaneously consider multiple travel attributes. In this sense, the key research problem lies in discovering individual criterion weights, that is, in basing the ranking of alternatives on actual user preferences rather than on predefined generic weights. At the same time, an additional research challenge lies in reconciling such individualized route ranking with broader sustainable mobility goals at the level of the transport system.

3.2. Multicriteria Route Choice Model

In the field of multicriteria decision-making, many methods have been developed that enable the ranking of alternatives based on multiple criteria. Moreover, scientific and professional literature contains numerous studies devoted to comparative analyses of multicriteria decision-making methods, either at a general level or in specific areas of application. Based on such analyses, a few guidelines have also been developed to assist in the selection of the most appropriate method. In those analyses, the alignment of a

method with the characteristics of the specific problem and the application domain is often highlighted as a key criterion [37–39].

The selection of an appropriate method is of central importance, since choosing an unsuitable method may lead to a model that is not fully adapted to the specifics of the problem under consideration. In addition, it is necessary to consider the fact that the end user of the system is not an expert in multicriteria decision-making. In this sense, the method should be as simple as possible to apply and should not require complex procedures for expressing or weighting preferences that are, in practice, understandable only to experts. An additional criterion in method selection relates to ease of implementation, interpretability, and operational applicability under real-world conditions, for example, in use by travellers.

Among the established multicriteria decision-making methods, the Analytic Hierarchy Process (AHP) is one of the most widely used approaches for determining criterion weights through pairwise comparisons. The Weighted Sum Method (WSM), in particular, provides a transparent and operationally simple ranking framework. For the purposes of the multimodal travel choice model developed in this research, the WSM and AHP approaches were considered. The choice of these methods is based on previous research reported in [12], in which ease of application, interpretability, and suitability for use by non-expert users, that is, travellers, were treated as key requirements.

3.2.1. AHP and the Limits of Explicit Weight Elicitation

To enable the ranking of alternatives within multicriteria decision-making, it is necessary to define the relative importance of individual criteria. For the ranking of multimodal routes, it is therefore not sufficient merely to identify the relevant travel attributes, but also to determine their relative weights. In classical MCDM approaches, these weights are often determined explicitly, for example, through pairwise comparison methods, among which the AHP is the best known [39,40]. However, such an approach may be problematic when the decision makers are non-expert users, such as travellers in the everyday context of multimodal route choice [12]. Recent studies have also highlighted methodological challenges associated with the statistical evaluation of stated preference surveys in transport contexts [41].

One of the best-known methods based on pairwise comparisons is the AHP [39]. As supporting evidence from previous research [12], the applicability of the AHP method was examined to assess the ability of users to explicitly evaluate the relative importance of individual criteria using Saaty's scale. The analysis was conducted on a sample of $N = 49$ respondents in the context of multimodal route-choice scenarios in the city of Zagreb.

The results of the analysis showed that a substantial proportion of respondents did not achieve a satisfactory level of consistency in their judgments. More specifically, for approximately 57% of respondents, the value of the consistency ratio (CR) exceeded the recommended threshold of 10% [42,43]. Importantly, these were not only slight departures from the recommended level, since a subset of respondents displayed considerably higher inconsistency, indicating that pairwise comparisons were particularly demanding for some participants. The distribution of the obtained CR values is shown in Figure 1.

These previously reported findings provide the methodological motivation for moving beyond explicit pairwise comparison toward an approach in which user preferences are discovered from evaluations of concrete route alternatives. When viewed in the broader context of the decision-making process, such inconsistency is not unexpected. In multimodal route choice, the decision maker, that is, the traveller, most often acts as a non-expert and is not accustomed to systematically assessing the relative importance of a larger number of criteria. In practice, travellers more often rely on their own travel experiences, previous

journeys, and established travel behaviour patterns, which together shape their preferences and their so-called travel “genome” [36].

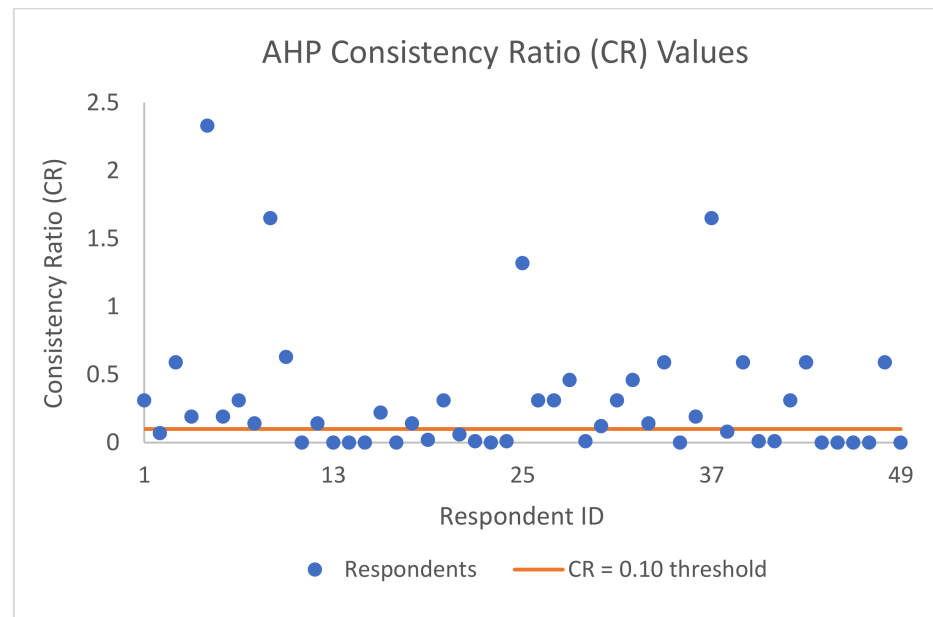


Figure 1. AHP Consistency Ratio (CR) values in the Zagreb analysis [12], with the 0.10 threshold indicated.

Under such circumstances, the decision-making process rarely follows the idealized model of full rationality. According to Simon, under conditions of limited information and cognitive capacity, decision makers in complex situations often seek a satisfactory, that is, sufficiently good solution, rather than a strictly optimal one [44]. When the complexity of multimodal travel is considered, including multiple modes, transfers, and time constraints, this mode of decision-making becomes particularly pronounced. In this sense, directly assessing the relative importance of travel criteria may represent a cognitively demanding task, especially in everyday and time-constrained decision situations, and may therefore result in greater inconsistency of judgments.

3.2.2. WSM as the Selected Route Choice Method

WSM is one of the mathematically simplest and most frequently applied multicriteria decision-making methods. It assumes additivity, according to which the overall utility of an alternative is determined as the sum of the weighted values of individual criteria [37,39]. This makes it possible to compare and rank alternatives under multicriteria decision-making conditions.

If a decision problem includes m alternatives evaluated according to n criteria, and if it is assumed that all criteria maximize the utility of an alternative, the decision model can be formally expressed as:

$$A_i = \sum_{j=1}^n w_j a_{ij} \quad (1)$$

where:

A_i	the overall value of the i -th alternative;
a_{ij}	the value of the j -th criterion for the i -th alternative;
w_j	the weight of the corresponding criterion;
i	the alternative index, $i = 1, 2, \dots, m$;
j	the criterion index, $j = 1, 2, \dots, n$;
m	the number of alternatives;
n	the number of criteria.

The optimal alternative is determined as the alternative with the highest overall value:

$$A_{opt} = \max_i A_i \quad (2)$$

where:

A_{opt}	the optimal alternative, that is, the alternative with the highest overall value;
A_i	the overall value of the i -th alternative obtained by applying the WSM model;
i	the alternative index ($i = 1, 2, \dots, m$).

The decision model can also be written in compact matrix form:

$$Alt = A \cdot W_c \quad (3)$$

where:

Alt	the vector of overall values of alternatives;
A	the matrix of alternative contributions by criteria (decision matrix);
W_c	the vector of criterion weights.

The optimal alternative is determined as the element of vector Alt with the highest value:

$$Alt(k) = \max(Alt) \quad (4)$$

Since WSM is based on additive aggregation, the criteria must be expressed on a comparable scale before the overall values of alternatives can be meaningfully calculated. When criteria are measured in different units, normalization is therefore a necessary step in order to avoid combining incomparable quantities. In this study, linear min–max normalization was applied to transform all criterion values into the interval $[0, 1]$, thereby ensuring comparability while preserving proportional relationships among values and improving the transparency of the model [37,45].

Within the presented framework, each alternative represents one multimodal route, while the criteria represent travel attributes describing the offered routes. Once the criterion weights have been determined, it becomes possible to calculate the overall value of each alternative and establish their ranking. In this way, WSM provides a simple and transparent framework for the operational implementation of a multimodal route-choice model.

3.3. Preference-Discovery-Based Model

The discovery of traveller preferences represents an important step in planning attractive and sustainable public transport [46,47], but also in the development of personalized journey-planning services that can respond more effectively to heterogeneous user needs and, in this way, provide the basis for the application of digital nudge mechanisms aimed at more sustainable mobility patterns [48].

In theory, preference discovery has traditionally been associated with the concept of revealed preferences, which originates in microeconomic theory and is linked to Samuelson's approach. The basic idea is that the actual preferences of consumers can be inferred from their observed behaviour, that is, from their choices among available alternatives [49,50].

The need for service personalization is closely related to the development of RSs, which today play an important role in information personalization and in supporting

users in the choice of products and services [34,51,52]. In this sense, the knowledge developed in the field of e-commerce and modern RSs can also be applied to the problem of multimodal route choice. Although personalized functionalities are increasingly present in passenger information systems, the literature still points to limitations in their implementation and to a lack of sufficiently elaborated frameworks for systematically shaping personalization [6,12,53]. Against this background, it is necessary to determine an appropriate methodology that will more reliably identify the actual needs and preferences of each user at the individual level.

Within the broader framework of recommender systems, different approaches to preference discovery have been developed, including collaborative methods, knowledge-based methods, and optimization-based approaches [54–56]. In the present context, preferences are most appropriately identified individually, for each user separately, rather than being derived from the preferences of a broader population group to which the user belongs.

One important class of recommendation algorithms is the model-based approach, in which a certain objective is optimized in order to improve the quality of recommendations. Such models most often focus on minimizing the error of predicting user desires, that is, preferences, or on improving the quality of recommendation ranking [55]. More recent studies also emphasize multi-objective recommendation optimization [56]. In this paper, an adapted optimization method for static system identification is applied, that is, an inverse decision-making model, using explicit user feedback on the offered alternatives for multimodal travel.

3.3.1. Optimization-Based Preference-Discovery Model

The general model of the decision-making process starts from a finite set of alternatives $\mathcal{A} = \{a_1, a_2, \dots, a_m\}$, a vector of criterion weights $W_c = \{w_j, j = 1, 2, \dots, n\}$ and a contribution matrix $A = [a_{ij}]$, which describes the contribution of each alternative with respect to each criterion. On the basis of these elements, the utility vector of alternatives Alt is determined, where the optimal alternative corresponds to the element with the highest utility value.

Unlike such a direct decision-making model, the present paper approaches the problem in the opposite direction. It starts from a known ranked or evaluated set of alternatives and a known contribution matrix A , while the aim is to estimate the vector of criterion weights W_c , that is, the implicit preferences of the user. In other words, instead of determining the utility of alternatives from previously known weights, the objective is to determine the relative importance of individual decision criteria from observed evaluations of alternatives.

Such an approach is defined as an inverse model of the decision-making process [57]. In the general case, the decision-making model can be written as [12]:

$$Alt(k) = \max(Alt), Alt = A \cdot W_c \quad (5)$$

where:

- A the matrix of contributions of each alternative with respect to each criterion (decision matrix);
- W_c the vector of criterion weights;
- Alt the vector of utility values of alternatives;
- $Alt(k)$ the element of vector Alt with the highest value.

Starting from the above model, the inverse problem is formulated as an optimization task of estimating the vector of criterion weights by minimizing the norm of the approxi-

mation error, that is, the deviation between the observed evaluations of alternatives and the values generated by the model:

$$\min_{W_c} \| Alt - A \cdot W_c \|^p \quad (6)$$

subject to:

$$\sum_{j=1}^n w_j = 1, \quad w_j \geq 0 \quad (7)$$

In this setting, vector Alt is determined on the basis of the respondent's evaluation, obtained by some method of ranking or rating the alternatives. Since in the general case this problem does not have a unique solution, the estimation of criterion weights is carried out by applying optimal estimation methods.

If $p = 2$ is chosen, the problem becomes a least-squares problem. In this way, the model seeks the vector of criterion weights that produces the smallest deviation between the user's evaluations of alternatives and the values predicted by the decision-making model. In that case, the optimization function takes the form:

$$\min_{W_c} \| Alt - A \cdot W_c \|^2 \quad (8)$$

that is, in expanded matrix form,

$$J = \| Alt - A \cdot W_c \|^2 = (Alt - A \cdot W_c)^T (Alt - A \cdot W_c) \quad (9)$$

Expanding the expression yields the standard quadratic form:

$$J = W_c^T Q W_c - 2W_c^T d + r \quad (10)$$

where:

$$Q = A^T A, d = A^T Alt, r = Alt^T Alt \quad (11)$$

A model formulated in this way enables numerical estimation of the vector of criterion weights under normalization and non-negativity constraints, where the resulting vector W_c represents an estimate of user preferences, that is, the relative importance of individual criteria in the decision-making process.

When using this approach, it is advisable to consider several multimodal route-choice scenarios, as this increases the accuracy of preference estimation. In the case of multiple scenarios, the model can be extended in form as follows:

$$Alt = \begin{bmatrix} Alt_1 \\ Alt_2 \\ \vdots \\ Alt_k \end{bmatrix}, A = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_k \end{bmatrix} \quad (12)$$

where:

Alt_i the estimated vector of alternative values in the i -th choice scenario;
 A_i the matrix of alternatives in the same scenario.

Under the assumption that the user does not change their fundamental preferences across the observed scenarios, such an integrated approach enables a more robust and more accurate estimation of the vector of criterion weights.

3.3.2. Extension Toward System-Aware Multimodal Route Re-Ranking

As described above, the proposed model is based on the estimation of individual criterion weights and the calculation of the overall values of alternatives using WSM. In this way, each alternative is assigned a value that reflects the degree of alignment between the offered route and the preferences of a particular user. However, personalization alone does not necessarily guarantee that the final recommended route will also be desirable from the broader perspective of the transport system. A route that is optimal for an individual may further burden already saturated parts of the network, have a less favourable environmental impact, or contribute less to the goals of sustainable mobility.

Transport systems are complex socio-technical systems encompassing infrastructure, traffic management systems, vehicles, and the human factor. In other words, transport functions as a highly complex system, and the decisions made by its actors directly affect other users. Since network capacities are limited, individual decisions, such as route choices made by drivers and passengers, directly affect the performance of the system. This interdependence makes transport a typical example of a system in which individual decisions must be analysed in relation to the overall rationality of the system.

This difference arises from the fact that individuals and the system often pursue conflicting goals. Whereas the individual is guided by minimizing personal cost, in terms of time or money, the system aims to reduce the total load on the network. This tension between individual and collective rationality can be interpreted through an analogy with concepts such as “selfish routing” and “The Tragedy of the Commons”, according to which individual decisions in shared-resource systems may lead to long-term systemic inefficiency [58,59]. In the transport safety domain, this tension is equally relevant: individually preferred routes may expose users and other road users to higher crash risk, particularly along segments with inadequate pedestrian or cycling infrastructure. Accordingly, journey planners do not function merely as information support for the user, but also as a mechanism for coordinating choices, one that may contribute to a balance of interests.

For this reason, the proposed preference-discovery-based model may be extended by a final system-aware corrective layer. The purpose of this extension is not to replace user preferences, but to adjust them toward solutions that are acceptable also from the perspective of the system. The final ranking thus results from a compromise between the individual optimum and the socio-system optimum. It is important to emphasize that this does not introduce a new inverse model, but only a re-ranking procedure applied to already personalized results.

The conceptual architecture of the proposed system-aware multimodal route recommender is shown in Figure 2. It illustrates the broader implementation context in which the preference-discovery-based model operates as a personalized ranking and re-ranking layer between the traveller, external routing services, and traffic/transport data infrastructure. The process starts when the user submits a route request to the recommender system, where the User Profile/Setup module stores the user profile created during the initial setup phase, including the initial set of preferences used for personalized route evaluation. The user preference re-discovery module is activated only when changes in travel behaviour or travel patterns are detected, enabling the system to update the preference structure and generate a revised set of user-specific criterion weights. Based on the route alternatives provided by external/global routing services, such as Google, Waze, or similar platforms, the recommender system first produces a personalized ranking of alternatives, meaning that the proposed system does not perform routing within the transport network itself, but acts as a preference-aware ranking and re-ranking layer applied to already generated candidate routes.

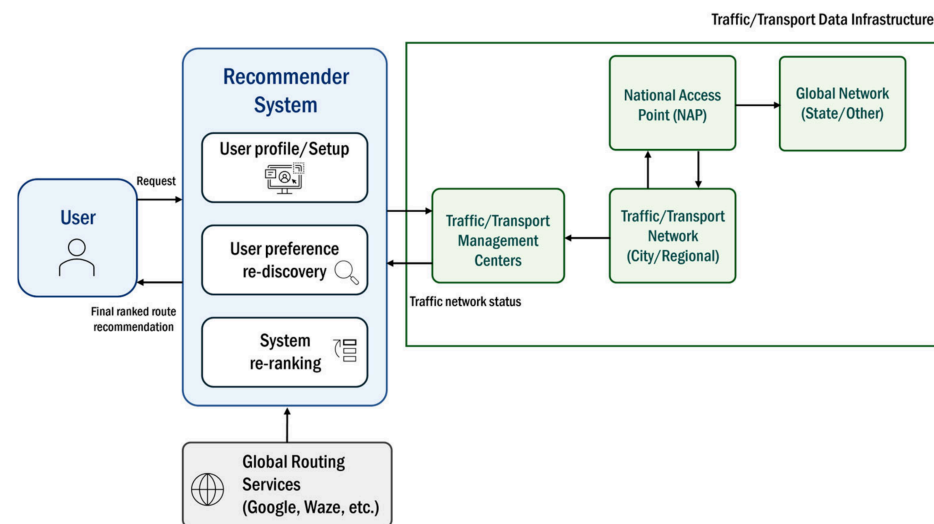


Figure 2. Conceptual architecture of the proposed system-aware multimodal route recommender.

Since the individually preferred route is not necessarily optimal from the perspective of the wider transport system, the architecture also includes a final system re-ranking layer. In this step, traffic/transport management centers and related data infrastructures provide input data for operationalizing system-level criteria, rather than directly replacing the user-based ranking. These inputs primarily support the assessment of network conditions and network impact, while other criteria, such as environmental impact and sustainable mode contribution, may be derived from route composition, emission factors, and data on the availability and use of sustainable transport modes. The National Access Point (NAP) is interpreted as a structured access point for transport-related data, connecting local and regional transport data with the wider national and cross-border transport data network. This is particularly relevant because multimodal travel information services are still often fragmented in terms of modal and geographical coverage, real-time data availability, and service quality. Therefore, open, harmonized, and interoperable transport data represent an important prerequisite for developing advanced and transparent MJP solutions [20]. In this way, the final recommendation is obtained by applying a system-aware re-ranking procedure to a narrower set of already acceptable alternatives, so that the final ranked route recommendation reflects a controlled compromise between individual preference alignment and broader transport system objectives, including network efficiency, environmental performance, and support for sustainable mobility.

For that purpose, an additional set of system criteria g is introduced:

$$G_s = \{g_1, g_2, g_3\} \quad (13)$$

where:

- g_1 network impact;
- g_2 environmental impact;
- g_3 sustainable mode contribution.

It should be noted that, depending on the availability of data and the goals of the transport authority, additional system-level criteria may be incorporated, such as road safety performance of the route segments, which would further enhance the societal relevance of the system-aware layer.

Criterion g_1 describes the extent to which a given route affects the load on the transport network. Higher desirability is assigned to routes that use less loaded segments of the network, whereas lower desirability is assigned to routes that further burden already

congested parts of the system. Criterion g_2 describes the environmental impact of the route, for example, through estimated emissions associated with the transport modes used. In this sense, routes with a lower environmental impact are assigned greater desirability. Criterion g_3 describes the extent to which a route includes sustainable transport modes, where greater desirability is assigned to routes that rely more on walking, cycling, and public transport.

After the overall values of alternatives are calculated based on the estimated vector of criterion weights W_c and the alternative contribution matrix A , a personalized ranking of alternatives is obtained from the perspective of a specific user. For clearer distinction from the final, system-corrected value, this vector may hereafter be denoted as $Alt^{(p)}$.

In order to take broader system objectives into account, an additional vector of system values of alternatives is introduced, determined on the basis of the matrix of system criteria G and the corresponding vector of weights β :

$$Sys = G \cdot \beta \quad (14)$$

where:

G the matrix of alternative contributions with respect to system criteria (decision matrix);

β the vector of system-criterion weights;

Sys the vector of system values of alternatives.

Accordingly, the final value of alternatives can be determined as a linear combination of the personalized and the system component:

$$Alt^{(f)} = (1 - \lambda)Alt^{(p)} + \lambda Sys \quad (15)$$

where:

$Alt^{(f)}$ the vector of final values of alternatives;

$Alt^{(p)}$ the vector of alternative values obtained from user preferences;

Sys the vector of system values of alternatives;

$\lambda \in [0, 1]$ the trade-off coefficient between the individual and system component.

If $\lambda = 0$, the final ranking of alternatives fully corresponds to the personalized order derived from user preferences. As the value of parameter λ increases, the influence of system criteria on the final ranking also increases. In this way, a controlled balance is established between user preferences and the broader goals of the transport system. It should be noted that β and λ are system parameters: they are not estimated from user ratings of alternatives but may be specified normatively or calibrated depending on system goals and the desired intensity of system correction.

However, in order to avoid situations in which system criteria lead to the recommendation of an alternative that is unacceptable to the user, it is proposed that the final system-aware layer should not be applied to the entire set of available alternatives, but only to a narrower set of alternatives that have already been assessed as sufficiently acceptable within the personalized model. In this sense, a *top-k* re-ranking procedure is introduced, whereby the set of the k highest-ranked alternatives is first determined on the basis of vector Alt :

$$T_k = Top - k(Alt) \quad (16)$$

After that, the system correction is applied only within set T_k , that is, only to alternatives satisfying the condition:

$$i \in T_k \quad (17)$$

Such an approach has several important methodological advantages. First, it respects the user, since the final recommendation is derived from a set of alternatives that are already aligned with their preferences. Second, it still leaves room for the inclusion of social and operational goals, since within this narrower set it becomes possible to favour an alternative that places less burden on the network, has a lower environmental impact, or contributes more strongly to sustainable mobility. Third, it reduces the risk that the system will recommend alternatives that may be formally desirable from the system perspective, but operationally unacceptable for a particular user.

In this way, the proposed model retains the basic WSM logic and its user-oriented core, while at the same time being extended by a final system layer enabling system-aware re-ranking of multimodal routes. Such a model structure is particularly suitable in the context of multimodal journey planning, where route recommendations should not be guided only by individual suitability, but also by broader goals of efficiency, sustainability, and balanced operation of the transport network.

The system criteria g can be operationalized using several complementary data sources. Network impact (g_1) may be estimated on the basis of data describing the current state and use of the transport network, including GTFS and GTFS Realtime data, operator data, traffic management data, and, where appropriate, additional sources such as crowdsourced traffic information or public bike-sharing data. These data may contribute to the assessment of congestion levels, network load, service availability, and other indicators relevant for estimating the operational impact of a given route. Environmental impact (g_2) may be evaluated using mode-specific emission factors and the modal composition of the route, while sustainable mode contribution (g_3) may be derived from the extent to which the route relies on walking, cycling, public transport, or other sustainable transport modes.

4. Case Study and Evaluation Design

4.1. Vienna Case Study

For the purposes of evaluating the proposed preference-estimation approach within a controlled methodological testbed, a travel case from the wider Vienna area was selected. The reason for this choice is the highly developed multimodal public transport system in that city. In addition, data from Statistik Austria confirm that everyday transport conditions in Vienna are characterized by a high level of public transport use and a relatively lower dependence on private cars [60]. Such a mobility pattern makes Vienna particularly suitable for examining the initial setup of user preferences within the empirical framework previously developed in [12].

4.2. Survey Design

For the purposes of the study, contact was established with institutions involved in research and development related to public transport in Vienna, and these institutions provided support in distributing an online questionnaire among users of the multimodal public transport system. After defining appropriate multimodal travel scenarios, an online questionnaire was developed [12]. The questionnaire and multimodal travel scenarios were adopted from the empirical design presented in [12]. The OD scenarios were defined within the wider urban area of Vienna, and Google Maps was used as the external routing service to generate candidate route alternatives. For each OD scenario, three multimodal route alternatives were selected, which was considered appropriate for the proof-of-concept design of the study and for keeping the respondent evaluation task manageable. The alternatives were reviewed to ensure that the final choice set included realistic and clearly distinguishable multimodal routes suitable for respondent evaluation. A total of 28 responses were collected, and after excluding incomplete and inconsistent questionnaires, a final sample

of $N = 21$ respondents was included in the analysis. Such a sample size is acceptable in this case because the aim of the study was not to generalize preferences at the population level, but rather to conduct a methodological proof-of-concept evaluation at the level of individual decision-making patterns. Nevertheless, it should be acknowledged that a larger and more heterogeneous sample would strengthen the external validity of the findings and is therefore recommended for future applications of the proposed methodology.

The questionnaire consisted of two parts: in the first part, respondents provided a general assessment of the importance of criteria, while in the second part, they evaluated each alternative within three multimodal travel scenarios, each comprising three offered routes. The empirical dataset is used solely as a controlled methodological testbed for evaluating the proposed individual-level preference-discovery methodology and is not intended to characterize mobility conditions or user behaviour in Vienna at any specific point in time.

4.3. Evaluation Procedure

For the purpose of evaluating the proposed preference-estimation approach, two in-sample procedures were carried out: a ranking-agreement analysis and a numerical error comparison. The evaluation procedure is shown in Figure 3. The evaluation procedures and related calculations were implemented in MATLAB (R2024a). In both cases, the user's vector of alternative ratings (Alt.-val.-rating) is taken as the reference vector and is compared with the vector of alternative values obtained on the basis of user-rated criterion values (Alt.-val.-crit.) and the vector of alternative values obtained on the basis of the preference-discovery model described in Section 3.3 (Alt.-val.-disc.). A common prerequisite for both procedures includes the selection of the final study sample, appropriate data preprocessing, and normalization of the input values, while also considering the user's perception of walking.

It should be emphasized that both procedures are conducted within the same methodological testbed. An out-of-sample procedure was not applied in the present proof-of-concept study. Therefore, the results are interpreted as descriptive in-sample agreement indicators, that is, goodness-of-fit measures between user-provided route evaluations and model-derived alternative values, rather than as evidence of out-of-sample predictive performance.

In the first evaluation procedure, a ranking-agreement analysis is employed. In this case, the accuracy of the ordering of alternatives is assessed as the fundamental output of the choice process. It is assumed that the ordering of alternatives provided by the user, that is, vector Alt.-val.-rating, is the reference ordering, against which the orderings obtained from vectors Alt.-val.-crit. and Alt.-val.-disc. are compared. The procedure is implemented by assigning penalty points to deviations from the reference ordering. If the user's best alternative, according to vector Alt.-val.-rating, is ranked in third place in the observed vectors Alt.-val.-crit. or Alt.-val.-disc., three penalty points are assigned; if it is ranked in second place, two penalty points are assigned; and if the user's second-best alternative is ranked in third place, one penalty point is assigned. Penalty points are summed up for each user across all three scenarios. The minimum total is zero, meaning that the ordering of alternatives fully matches the ordering given by the user, while the maximum total is nine, corresponding to the case in which the best alternative is ranked only in third place in all three scenarios.

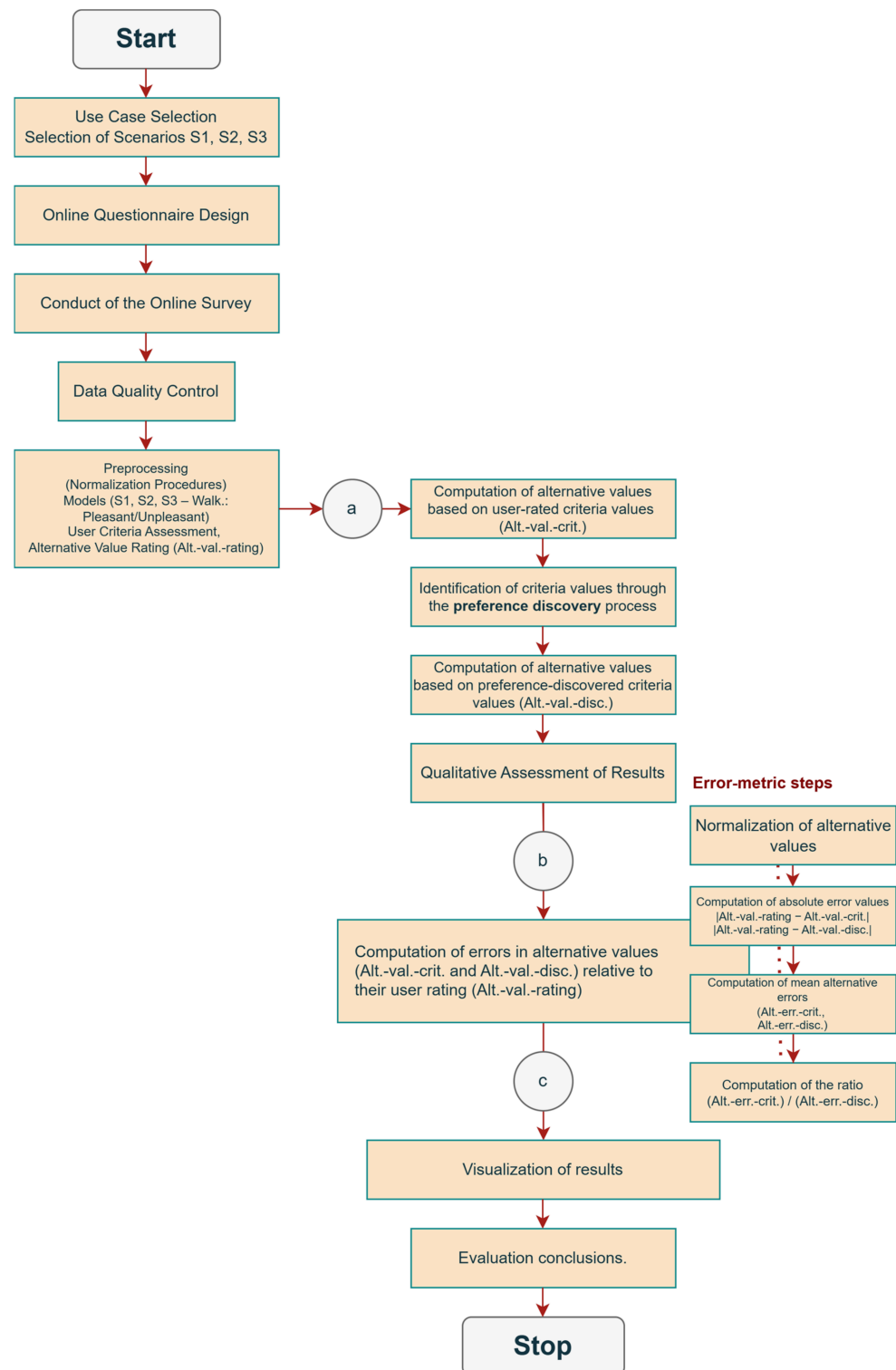


Figure 3. Evaluation procedure for the preference-estimation approach [12].

In the second evaluation procedure, a numerical error comparison is applied. The reference vector is again the user's alternative rating vector (Alt.-val.-rating), which is compared with the vectors of alternative values obtained from the user's criterion ratings (Alt.-val.-crit.) and from the preference-discovery model (Alt.-val.-disc.). Unlike the ranking-agreement analysis, this procedure does not examine the ordering of alternatives itself, but the numerical deviations of their values. As a first step, the errors of the calculated alternative values, namely Alt.-val.-crit. and Alt.-val.-disc., are computed relative to the

normalized user ratings (Alt.-val.-rating). Because of the nature of the error, its absolute value is taken. After that, the mean value of the absolute errors across all three scenarios is calculated for each user. As an in-sample measure of agreement between user-provided route evaluations and model-derived alternative values, the ratio of the mean absolute errors obtained for Alt.-val.-disc. and Alt.-val.-crit. is calculated. Finally, to provide clearer insight into the results, an appropriate visualization of the results is also carried out.

5. Results

The results indicate that, within the available methodological testbed, the preference-discovery-based model achieved closer in-sample agreement with user-provided route evaluations than the model based on user-rated criterion values. This was observed in two evaluation procedures: the ranking-agreement analysis, where the preference-discovery-based model produced rankings with fewer penalty points, and the numerical error comparison, where it showed lower in-sample reconstruction errors. Overall, the findings indicate that the preference-discovery method can represent user evaluations more closely than explicitly specified criterion weights, thereby highlighting the limitations of explicit user-defined criterion weighting in the initial preference setup.

5.1. Results of the Ranking-Agreement Analysis

The ranking-agreement analysis shows the extent to which the orderings of alternatives obtained by the observed models coincide with the reference ordering provided by the user. Figure 4 presents the ranking agreement results as the total number of penalty points across all three scenarios, for each respondent and for both compared multimodal route-choice models: the user-rated criterion model and the preference-discovery-based model. The figure indicates that the preference-discovery-based model achieved equal or better ranking agreement than the user-rated criterion model for all respondents, as reflected by its lower or equal number of total penalty points.

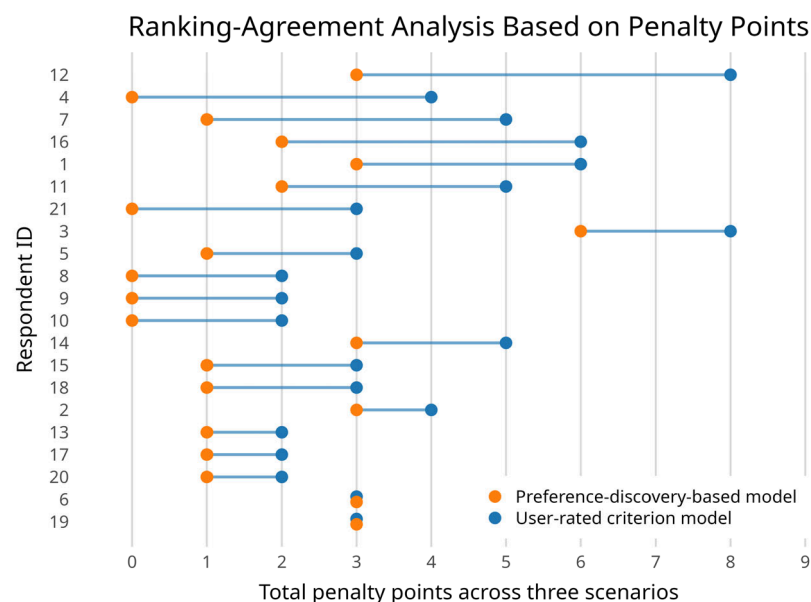


Figure 4. Results of the ranking-agreement analysis based on data from [12].

Respondents are ordered according to the gap between the user-rated criterion model and the preference-discovery-based model. For respondents 6 and 19, both models obtained the same number of penalty points.

Additional analysis of the penalty-point ratios showed that there was not a single respondent for whom the model based on user-rated criterion values achieved a better match with the reference ordering than the preference-discovery-based model. Only for two respondents did both models produce the same result, while for five respondents the preference-discovery-based model fully matched the user-provided ordering of alternatives. If a value of 0.8 is taken as an author-defined reference value for interpreting the penalty-point ratio, a more favourable ratio for the preference-discovery-based model is observed in 19/21 cases (90.5%). This value is not interpreted as a statistical threshold or as evidence of predictive superiority, but as a descriptive in-sample indicator within the available methodological testbed.

5.2. Numerical Error Comparison

The numerical error comparison shows that, for the majority of users, the in-sample reconstruction error is lower in the case of the preference-discovery-based model than in the case of the model based on user-rated criterion values. This pattern appears in all three observed scenarios, although with varying intensity. Lower error values for the preference-discovery-based model were obtained for 15/21 users (71.43%) in Scenario 1, 14/21 users (66.67%) in Scenario 2, and 19/21 users (90.48%) in Scenario 3.

In other words, the numerical error comparison indicates that, in most cases within the available methodological testbed, the preference-discovery-based model more closely reconstructs the user's numerical evaluation of the offered routes. An integrated presentation of the results across all three scenarios is provided in Figure 5, which shows the relationship between the errors in estimating alternative values. For 18/21 users (85.71%), lower in-sample reconstruction errors were obtained with the preference-discovery-based model.

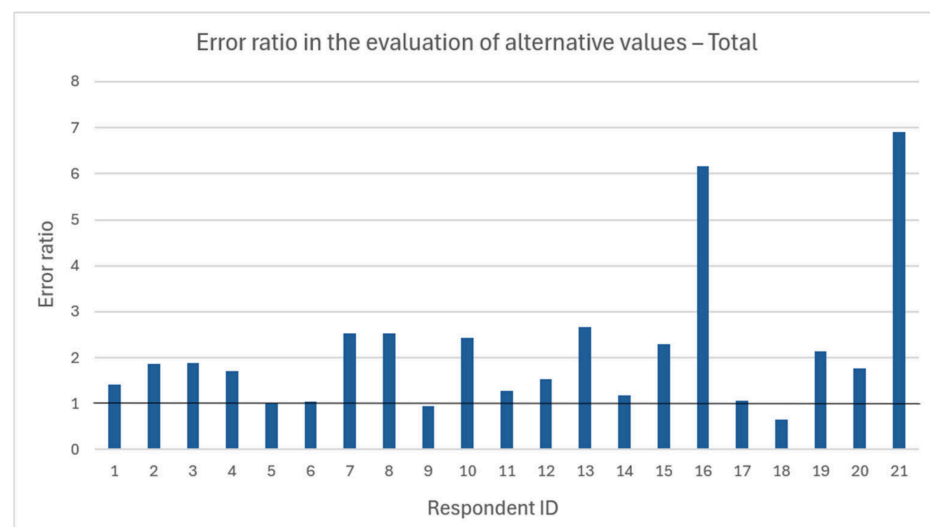


Figure 5. Error ratio for alternative value estimation across all three scenarios [12].

Overall, the general pattern is observed in the numerical error comparison, namely that the preference-discovery-based model shows lower in-sample reconstruction errors for most users within the available methodological testbed. This result supports the feasibility of estimating interpretable user-specific preference weights from route evaluations in the initial preference setup stage, while not representing out-of-sample predictive validation.

6. Discussion and Conclusions

The results of the two in-sample evaluation procedures indicate that, within the available methodological testbed, the preference-discovery-based model achieved closer

agreement with user-provided route evaluations than the user-rated criterion model. In the ranking-agreement analysis, there was not a single user for whom the model based on explicitly rated criteria achieved a better match with the reference ordering than the preference-discovery-based model. In the numerical error comparison, the preference-discovery-based model also produced lower in-sample reconstruction errors for most users. These findings should be interpreted as descriptive goodness-of-fit results for the initial preference setup; at the same time, they indicate that the preference-discovery method can represent user evaluations more closely than explicitly stated criterion weights, thereby highlighting one of the main motivations of the study: the limitations of explicit user-defined criterion weighting in current multimodal journey planners.

The obtained results are interpreted as descriptive in-sample agreement results within the initial preference setup. Already at the theoretical level, the paper starts from the premise that multimodal route choice is a complex multicriteria decision-making problem in which users evaluate trade-offs among several travel attributes, while generally acting as non-experts. In this sense, it is difficult to expect that they will be able to assess the importance of individual criteria in a stable and consistent way directly. It is precisely for this reason that the model based on the WSM was selected in the paper as the most appropriate framework, because it enables a well-structured formulation of the problem, explicitly defines alternatives, criteria, and their contributions, and at the same time remains sufficiently simple and interpretable for operational application. By contrast, AHP, both in previous research and in the dissertation on which this work builds, has shown significant limitations in the case of non-expert users, particularly because of the difficulty of consistently expressing the relative importance of criteria.

An important contribution of the paper also lies in the preference discovery methodology itself. The paper starts from the assumption that there is a direct relationship between the user's choice of concrete alternatives and the corresponding set of preferences relevant to the activity under consideration. Preferences are therefore not determined directly but are estimated implicitly on the basis of the user's evaluations of the offered alternatives. Mathematically, this approach is formulated as an optimization problem in which the vector of criterion weights is determined by minimizing the p -th norm of the approximation error of the decision-making model, whereby in the case of the quadratic norm the problem reduces to the least-squares method. Such a formulation enabled the estimation of interpretable user-specific criterion weights and achieved closer in-sample agreement with user evaluations within the available methodological testbed. The results support the relevance of identifying user preferences individually, for each user separately, rather than deriving them from average patterns of a broader population.

The findings also have broader implications for the role of MJPs in smart mobility systems, particularly as digital decision-support tools within smart cities. Journey planners should therefore serve not only personalization purposes, but also the alignment of individual and system rationality. This implies a transparent and verifiable logic of alternative ranking, which is why white-box approaches have value compared with AI-based methods (grey to black-box models), since they make recommendations both personalized and interpretable [61].

If implicitly estimated preferences achieve closer agreement with user-provided route evaluations than explicitly stated criterion weights, then personalization in such systems should not be reduced to predefined filters or generic priorities but should instead be based on modelling actual individual decision-making patterns. In this sense, the proposed approach is relevant not only as a method for estimating user criterion weights, but also as a broader methodological basis for the development of transparent, preference-aware recommender systems in the field of multimodal mobility. The conceptual extension toward

system-aware re-ranking additionally shows that the user-oriented core of the model can be upgraded by a final system layer, thereby creating space for the inclusion of broader transport-system goals. At the present stage, however, this part of the study should be interpreted as a methodological extension and an illustration of the mechanism, rather than as a separately empirically validated finding.

At the same time, the paper also has clear limitations. The empirical dataset was used as a controlled methodological testbed for evaluating the preference-discovery methodology, and not for generalizing patterns of mobility or user behaviour in Vienna. The sample is relatively small and suitable primarily for a methodological proof-of-concept evaluation at the level of individual decision-making patterns. Given the study's focus on methodological proof-of-concept and individual-level preference estimation, the sample size of $N = 21$ was sufficient for the initial evaluation within the established methodological framework. The intentionally limited set of criteria also reflects the proof-of-concept and initial preference setup focus of the study, while additional inconvenience-related factors should be considered in future operational extensions of the model. Moreover, the estimated weights were evaluated within the same methodological testbed; an out-of-sample procedure on new route-choice scenarios was not applied in the present proof-of-concept study. Therefore, the reported results should be interpreted as descriptive in-sample agreement indicators.

Future research should therefore include out-of-sample evaluation on new route-choice scenarios and a broader statistical evaluation based on a larger and more diverse sample, in order to assess the stability, transferability and broader generalizability of the estimated preference profiles. Future research may also focus on the operational elaboration of system criteria under real implementation conditions, as well as on further extension of the model toward intelligent and adaptive preference modelling approaches, including uncertainty-aware representations, AI-assisted estimation procedures, possible extensions of the alternative evaluation framework, and the integration of safety-related criteria reflecting infrastructure quality and crash risk along multimodal route segments. Future work may also examine the integration of gamification mechanisms into MJPs, since recent studies [62,63] suggest that gamified digital tools can support behavioural change and user engagement in sustainable urban mobility. Their effective application in journey planning still requires further investigation regarding incentive design, user acceptance, and the persistence of behavioural effects over time.

Author Contributions: Conceptualization, B.M. and K.V.; methodology, B.M., K.V. and J.Č.; validation, B.M. and K.V.; writing—original draft preparation, B.M.; visualization, B.M.; supervision, M.Š. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed at the corresponding author.

Acknowledgments: The authors would like to gratefully acknowledge the support and cooperation of colleagues and students from Technische Universität Wien, particularly from the Research Center of Transport Planning and Traffic Engineering, in relation to the part of the research conducted for the wider urban area of Vienna.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Van Nes, R.; Bovy, P.H.L. Multimodal Traveling and Its Impact on Urban Transit Network Design. *J. Adv. Transp.* **2004**, *38*, 225–241. [CrossRef]
2. Kosmidis, I.; Müller-Eie, D. Measuring Perceived Transfer Inconvenience for Multimodal Commute Trips Combining Bicycles and Public Transport. *Travel Behav. Soc.* **2026**, *43*, 101167. [CrossRef]
3. Cascajo, R.; Lopez, E.; Herrero, F.; Monzon, A. User Perception of Transfers in Multimodal Urban Trips: A Qualitative Study. *Int. J. Sustain. Transp.* **2019**, *13*, 393–406. [CrossRef]
4. OECD. *Transport Strategies for Net-Zero Systems by Design*; OECD Publishing: Paris, France, 2021.
5. Hou, N.; Shollock, B.; Petzoldt, T.; M’Hallah, R. Qualitative Insights into Travel Behavior Change from Using Private Cars to Shared Cars. *Int. J. Sustain. Transp.* **2025**, *19*, 262–276. [CrossRef]
6. van Ardenne, M.T.; Cebecauer, M.; Cats, O.; Ma, Z. Personalised Passenger Information Systems in Public Transport: A Review and a 5-Level Personalisation Taxonomy. *Transp. Rev.* **2025**, *45*, 1016–1047. [CrossRef]
7. Mandžuka, B.; Jurčević, M.; Vidović, K. Mobility-as-a-Service and the Future of Urban Mobility: A Human-Centered Approach to Multimodal Journey Planning. In *New Technologies, Development and Application VI; Lecture Notes in Networks and Systems*; Springer: Cham, Switzerland, 2023; Volume 687, pp. 729–738. [CrossRef]
8. Sandu, A.; Cotfas, L.A.; Stănescu, A.; Delcea, C. Guiding Urban Decision-Making: A Study on Recommender Systems in Smart Cities. *Electronics* **2024**, *13*, 2151. [CrossRef]
9. Allam, Z.; Sharifi, A. Research Structure and Trends of Smart Urban Mobility. *Smart Cities* **2022**, *5*, 539–561. [CrossRef]
10. Vujić, M.; Mandžuka, S.; Gregurić, M. Pilot Implementation of Public Transport Priority in the City of Zagreb. *Promet-Traffic Transp.* **2015**, *27*, 257–265. [CrossRef]
11. Hsieh, F.-S. Emerging Research Issues and Directions on MaaS, Sustainability and Shared Mobility in Smart Cities with Multi-Modal Transport Systems. *Appl. Sci.* **2025**, *15*, 5709. [CrossRef]
12. Mandžuka, B. Model Izboru Multimodalnoga Putovanja Temeljen Na Otkrivanju Putničkih Preferencija. Doctoral Dissertation, University of Zagreb, Zagreb, Croatia, 2022.
13. Esztergár-Kiss, D. Framework of Aspects for the Evaluation of Multimodal Journey Planners. *Sustainability* **2019**, *11*, 4960. [CrossRef]
14. Malichová, E.; Straka, M.; Buzna, L.; Scandolari, D.; Scrocca, M.; Comerio, M. Study of Travellers’ Preferences towards Travel Offer Categories and Incentives in the Journey Planning Context. *PLoS ONE* **2023**, *18*, e0284844. [CrossRef] [PubMed]
15. Ranjan, R.; Sinha, S. A Systematic Review of Mode Choice Behavior in Urban Transportation with Emphasis on Individual Preferences and Influencing Factor. *Discov. Cities* **2025**, *2*, 98. [CrossRef]
16. Alharthi, S.; El-Saddik, A. Personalization in Smart Urban Environments: A Taxonomy and Survey of Recommender Systems. *Comput. Urban Sci.* **2026**, *6*, 8. [CrossRef]
17. Salanova Grau, J.M.; Nikiforiadis, A.; Tzenos, P.; Tsavdari, D.; Ayfantopoulou, G. Development of an Individualized Multimodal Trip Planner for a MaaS System. *Sustain. Futures* **2025**, *9*, 100498. [CrossRef]
18. El Bouhissi, H.; Hanne, T.; Madadi, M. Recommender Systems for Multimodal Transportation in Smart Sustainable Cities. *Sustainability* **2025**, *17*, 10810. [CrossRef]
19. Vidovic, K.; Mandzuka, S.; Sostaric, M. Data Quality Within National Access Point for Provision of Multimodal Travel Information within European Union. In *Proceedings of the Proceedings Elmar—International Symposium Electronics in Marine*; IEEE: Zadar, Croatia, 2019; pp. 15–19.
20. Mandžuka, B.; Vidović, K.; Vujić, M.; Alexopoulos, C. The Importance Of Open Data Accessibility For Multimodal Travel Improvement. *Interdiscip. Descr. Complex Syst. INDECS* **2022**, *20*, 136–148. [CrossRef]
21. European Commission. *Sustainable and Smart Mobility Strategy: Putting European Transport on Track for the Future*; European Commission: Brussels, Belgium, 2020.
22. European Commission Multimodal Travel Information. Available online: https://transport.ec.europa.eu/transport-themes/smart-mobility/road/its-directive-and-action-plan/multimodal-travel-information_en? (accessed on 21 April 2026).
23. Spickermann, A.; Grienitz, V.; Von Der Gracht, H.A. Heading towards a Multimodal City of the Future?: Multi-Stakeholder Scenarios for Urban Mobility. *Technol. Forecast. Soc. Change* **2014**, *89*, 201–221. [CrossRef]
24. Hodson, M.; Marvin, S. Can Cities Shape Socio-Technical Transitions and How Would We Know If They Were? *Res. Policy* **2010**, *39*, 477–485. [CrossRef]
25. Yu, Y.; Li, D.; Han, B.; Zhang, Q.; Huang, Y.; Yang, R. Enhancing Urban Mobility: A Multi-Modal Travel Plan Recommendation Framework Integrating the Influences of Temporal Characteristics and Candidate Sets. *Inf. Sci.* **2025**, *709*, 122042. [CrossRef]
26. Bıyık, C.; Abareshi, A.; Paz, A.; Ruiz, R.A.; Battarra, R.; Rogers, C.D.F.; Lizarraga, C. Smart Mobility Adoption: A Review of the Literature. *J. Open Innov. Technol. Mark. Complex.* **2021**, *7*, 146. [CrossRef]
27. Quijano-Sánchez, L.; Cantador, I.; Cortés-Cediel, M.E.; Gil, O. Recommender Systems for Smart Cities. *Inf. Syst.* **2020**, *92*, 101545. [CrossRef]

28. Liu, S.; Chen, S.; Yu, D.; Zhu, Y.; Yao, E.; Hao, M. Personalized Guidance Information and Travel Choice Behavior During Metro Service Disruptions: Evidence from Beijing, China. *Urban Sci.* **2025**, *9*, 546. [\[CrossRef\]](#)
29. Zimmermann, S.; Schulz, T.; Hein, A.; Gewald, H.; Krcmar, H. Motivating Change in Commuters' Mobility Behaviour: Digital Nudging for Public Transportation Use. *J. Decis. Syst.* **2024**, *33*, 79–105. [\[CrossRef\]](#)
30. Ćelić, J.; Mandžuka, B.; Tomas, V.; Tadić, F. Driver-Centric Urban Route Planning: Smart Search for Parking. *Sustainability* **2024**, *16*, 856. [\[CrossRef\]](#)
31. Lu, Q.C.; Zuo, X.Y.; Chen, C.; Dong, Z.; Xu, P.C.; Li, J. The Heterogeneity of Travel Mode Choice Behavior under Unplanned Metro Service Disruptions. *Transp. Res. D Transp. Environ.* **2025**, *142*, 104683. [\[CrossRef\]](#)
32. Puan, O.C.; Hassan, Y.A.H.; Mashros, N.; Kriswardhana, W.; Esztergár-Kiss, D. Identifying Heterogeneity in University Students' Transport Mode Choice. In *Proceedings of the IOP Conference Series: Earth and Environmental Science*; IOP Publishing: Bristol, UK, 2024; Volume 1294, p. 012027.
33. Di Martino, S.; Rossi, S. An Architecture for a Mobility Recommender System in Smart Cities. *Procedia Comput. Sci.* **2016**, *98*, 425–430. [\[CrossRef\]](#)
34. Wu, F.; Lyu, C.; Liu, Y. A Personalized Recommendation System for Multi-Modal Transportation Systems. *Multimodal Transp.* **2022**, *1*, 100016. [\[CrossRef\]](#)
35. Alkhabbas, F.; De Sanctis, M.; Bucchiarone, A.; Cicchetti, A.; Spalazzese, R.; Davidsson, P.; Iovino, L. ROUTE: A Framework for Customizable Smart Mobility Planners. In *Proceedings of the IEEE 19th International Conference on Software Architecture, ICSA 2022*; Institute of Electrical and Electronics Engineers Inc.: New York, NY, USA, 2022; pp. 169–180.
36. Goulias, K.G.; Davis, A.W.; McBride, E.C. Introduction and the Genome of Travel Behavior. In *Mapping the Travel Behavior Genome*; Elsevier: Amsterdam, The Netherlands, 2020; pp. 1–14. [\[CrossRef\]](#)
37. Malefaki, S.; Markatos, D.; Filippatos, A.; Pantelakis, S. A Comparative Analysis of Multi-Criteria Decision-Making Methods and Normalization Techniques in Holistic Sustainability Assessment for Engineering Applications. *Aerospace* **2025**, *12*, 100. [\[CrossRef\]](#)
38. Makridis, M.; Rakha, H.A.; Fontaras, G.; Solecka, K.; Kiciński, M. A Multi-Criteria Evaluation of Applications Supporting Public Transport Users. *Energies* **2022**, *15*, 3493. [\[CrossRef\]](#)
39. Triantaphyllou, E. *Multi-Criteria Decision Making Methods: A Comparative Study; Applied Optimization*; Springer: Boston, MA, USA, 2000; Volume 44, ISBN 978-1-4419-4838-0.
40. Saaty, T.L. How to Make a Decision: The Analytic Hierarchy Process. *Interfaces* **1994**, *24*, 19–43. [\[CrossRef\]](#)
41. Sipos, T. Evaluation of Stated Preference Surveys with Statistical Methods. *Promet-Traffic Transp.* **2023**, *35*, 655–661. [\[CrossRef\]](#)
42. Saaty, T.L.V.L.G. *The Logic of Priorities*; RWS Publications: Pittsburgh, PA, USA, 1991.
43. Waris, M.; Panigrahi, S.; Mengal, A.; Soomro, M.I.; Mirjat, N.H.; Ullah, M.; Azlan, Z.S.; Khan, A. An Application of Analytic Hierarchy Process (AHP) for Sustainable Procurement of Construction Equipment: Multicriteria-Based Decision Framework for Malaysia. *Math. Probl. Eng.* **2019**, *2019*, 6391431. [\[CrossRef\]](#)
44. Simon, H.A. A Behavioral Model of Rational Choice. *Q. J. Econ.* **1955**, *69*, 99–118. [\[CrossRef\]](#)
45. Pavlicic, D.M. Normalisation Effects the Results of MADM Methods. *Yugosl. J. Oper. Res.* **2001**, *11*, 251–265.
46. Khademi-Vidra, A.; Nemecz, G.; Mária Bakos, I. Satisfaction Measurement in the Sustainable Public Transport of Budapest. *Transp. Res. Interdiscip. Perspect.* **2024**, *23*, 100989. [\[CrossRef\]](#)
47. Göransson, J.; Andersson, H. Factors That Make Public Transport Systems Attractive: A Review of Travel Preferences and Travel Mode Choices. *Eur. Transp. Res. Rev.* **2023**, *15*, 32. [\[CrossRef\]](#)
48. Dalecke, S.; Karlsen, R. Designing Dynamic and Personalized Nudges. In *Proceedings of the 10th International Conference on Web Intelligence; Mining and Semantics*, Biarritz, France, 30 June–3 July 2020; Association for Computing Machinery: New York, NY, USA, 2020; pp. 139–148.
49. Samuelson, P.A. A Note on the Pure Theory of Consumer's Behaviour. *Economica* **1938**, *5*, 61–71. [\[CrossRef\]](#)
50. Samuelson, P.A. Consumption Theory in Terms of Revealed Preference. *Economica* **1948**, *15*, 243–253. [\[CrossRef\]](#)
51. Adomavicius, G.; Tuzhilin, A. Toward the Next Generation of Recommender Systems: A Survey of the State-of-the-Art and Possible Extensions. *IEEE Trans. Knowl. Data Eng.* **2005**, *17*, 734–749. [\[CrossRef\]](#)
52. Zhou, H.; Xiong, F.; Chen, H.A.; Zhou, H.; Xiong, F.; Chen, H. A Comprehensive Survey of Recommender Systems Based on Deep Learning. *Appl. Sci.* **2023**, *13*, 11378. [\[CrossRef\]](#)
53. Esztergár-Kiss, D.; Csiszár, C. Evaluation of Multimodal Journey Planners and Definition of Service Levels. *Int. J. Intell. Transp. Syst. Res.* **2015**, *3*, 154–165. [\[CrossRef\]](#)
54. Najmani, K.; El Habib, B.; Sael, N.; Zellou, A. A Comparative Study on Recommender Systems Approaches. In *Proceedings of the 4th International Conference on Big Data and Internet of Things*; Association for Computing Machinery: New York, NY, USA, 2019; pp. 1–5.
55. Yi, P.; Li, C.; Yang, C.; Chen, M. An Optimization Method for Recommendation System Based on User Implicit Behavior. In *Proceedings of the 5th International Conference on Instrumentation and Measurement, Computer, Communication, and Control*, IMCCC 2015, Qinhuangdao, China, 18–20 September 2015; pp. 1537–1540.

56. Zheng, Y.; David, W. Multi-Objective Recommendations: A Tutorial. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*; Association for Computing Machinery: New York, NY, USA, 2021.
57. Shinohara, M. Inverse Problems in AHP. In *Proceedings of the International Symposium of the Analytic Hierarchy Process 2014*; Creative Decision Foundation: Washington, DC, USA, 2014; pp. 1–5.
58. Roughgarden, T.; Tardos, É. How Bad Is Selfish Routing? *J. ACM* **2002**, *49*, 236–259. [[CrossRef](#)]
59. Hardin, G. The Tragedy of the Commons. *Science* **1968**, *162*, 1243–1248. [[CrossRef](#)] [[PubMed](#)]
60. Umweltbedingungen, Umweltverhalten 2023: Ergebnisse Des Mikrozensus; Wien, 2024. Available online: <https://www.statistik.at/services/tools/serviceangebote/publikationen/detail/2061> (accessed on 20 March 2026).
61. Hashemi, M.; Darejeh, A.; Cruz, F. Understanding User Preferences in Explainable Artificial Intelligence: A Mapping Function Proposal. *ACM Trans. Intell. Syst. Technol.* **2025**, *16*, 1–37. [[CrossRef](#)]
62. Amaddeo, E.; Bergantino, A.S.; Gardelli, A.; Intini, M.; Shah, S.M.A. Gamification as Technological Innovation in Sustainable Urban Mobility: Evidence from a Digital Field Experiment. *Case Stud. Transp. Policy* **2026**, *24*, 101791. [[CrossRef](#)]
63. Wang, W.; Gan, H.; Wang, X.; Lu, H.; Huang, Y. Initiatives and Challenges in Using Gamification in Transportation: A Systematic Mapping. *Eur. Transp. Res. Rev.* **2022**, *14*, 41. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.